

# Microstructure Signals vs Executable Alpha

A three-study quantitative-research case on BTCUSDT perpetual futures

**Research question: Can short-horizon market-microstructure predictability survive out-of-sample testing and realistic execution constraints?**

## Abstract

This project studies static top-of-book imbalance, aggressive taker trade flow, and execution economics through a sequence of pre-specified gates. Study I finds a stable positive relationship between book imbalance and future 10-minute mid-price returns. A frozen logistic model retains weak untouched-test ranking power (AUC 0.5103), but conservative taker execution fails independent economic confirmation. Study II adds aggressive trade-flow variables and improves pre-OOS AUC, while rejecting the expected continuation sign: aggressive buying is followed by short-horizon reversal. Study III replicates that reversal on fresh 2025 data, but the signal is not robustly incremental to simple 5-minute price mean reversion. The final 2025 OOS therefore remains closed.

**Central result: statistical predictability, predictive generalization, and executable alpha are distinct claims. The project found reproducible information, but did not establish a profitable production strategy.**

## Key results

| Stage                   | Metric                         | Result            |
|-------------------------|--------------------------------|-------------------|
| Study I static book     | 10m validation beta            | +0.6647 bps/unit  |
| Study I model           | Untouched Jan-Mar 2024 AUC     | 0.5103            |
| Economic validation     | December 0-cost net mean       | -0.3427 bps/trade |
| Study II flow extension | December Delta AUC             | +0.0060           |
| Study III 2025          | Jan-Mar / Apr-Jun reversal AUC | 0.5210 / 0.5151   |
| Study III gate          | Incremental vs price reversal  | Failed            |

The research deliberately stops after failed gates rather than escalating model complexity post hoc.

## 1. Research design and controls

The work was designed as a quantitative-research case study rather than a trading-bot demonstration. The core sequence was: **hypothesis -> data integrity -> descriptive inference -> chronological prediction -> economic validation -> independent extension**.

### Controls added before trusting the full-data result

| Risk                         | Control                                                      |
|------------------------------|--------------------------------------------------------------|
| Overlapping forward returns  | HAC/Newey-West covariance plus non-overlapping samples       |
| Multiple comparisons         | Benjamini-Hochberg FDR                                       |
| Time-series leakage          | Chronological periods; no random split                       |
| Missing minute rows          | Complete clock grid; exact t+h endpoints only                |
| Target mechanical dependence | Future mid-price used instead of future microprice           |
| Boundary leakage             | Labels crossing train/validation boundaries purged           |
| Test-set overuse             | Explicit gates before test unlock                            |
| Backtest overfitting         | Threshold selection and cost grid frozen before confirmation |

A methodological audit also identified that microprice premium is essentially a deterministic interaction between close spread and close book imbalance. It was retained as a descriptive composite but excluded from the primary logistic feature matrix.

### Data

Study I used a public one-minute BTCUSDT perpetual book-ticker feature dataset with 460,265 observed rows and 99.726% minute-grid completeness. Studies II and III used official Binance USD-M one-minute kline archives, including total volume, quote volume, trade count, taker-buy base volume and taker-buy quote volume. The downloader verifies Binance SHA-256 checksum companion files.

### Terminology

**Book imbalance** is a state variable at the best bid/ask. **Trade-flow imbalance (TFI)** is derived from aggressive buyer/seller volume. It is intentionally not labelled full order-flow imbalance (OFI), because OFI also includes limit-order additions and cancellations.

## 2. Study I - Static top-of-book state

The primary hypothesis was that a bid-heavy top of book contains information about future mid-price direction. For a 10-minute horizon, close book imbalance had a positive coefficient in every pre-test month. Development and validation aggregate effects were +0.5255 and +0.6647 bps per unit imbalance, respectively, with dependence-aware inference and FDR support.

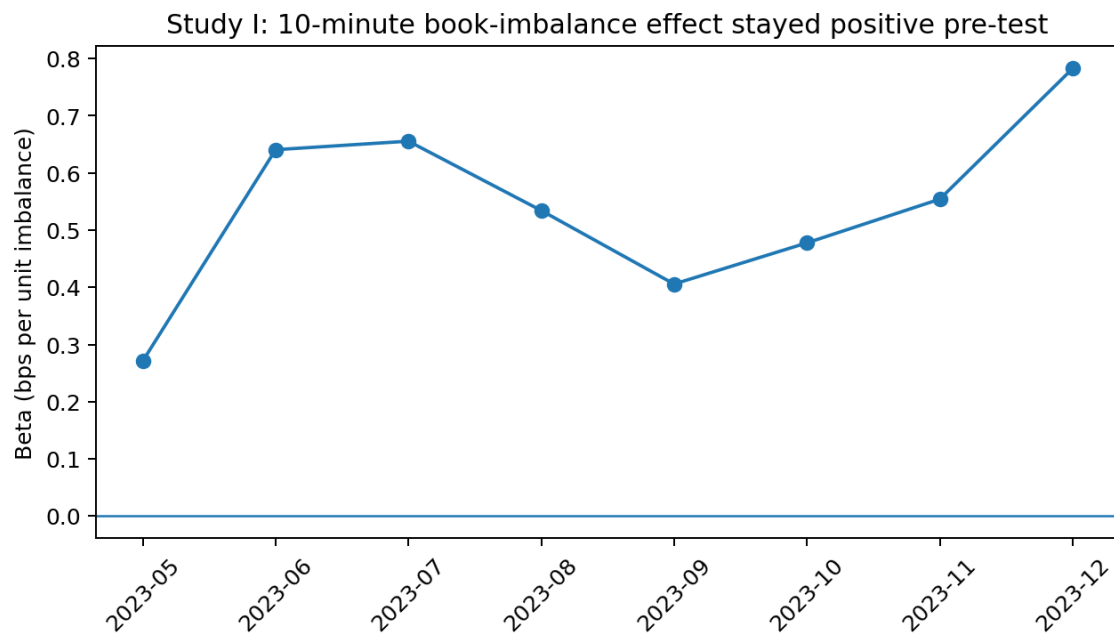


Figure 1. Monthly 10-minute close book-imbalance beta, May-December 2023.

The baseline logistic model was frozen before the final test. Performance declined chronologically but remained above chance: development AUC 0.5232, validation 0.5133, and untouched Jan-Mar 2024 test 0.5103. This supports only a narrow claim of weak ranking information; it does not establish a strong calibrated probability model.

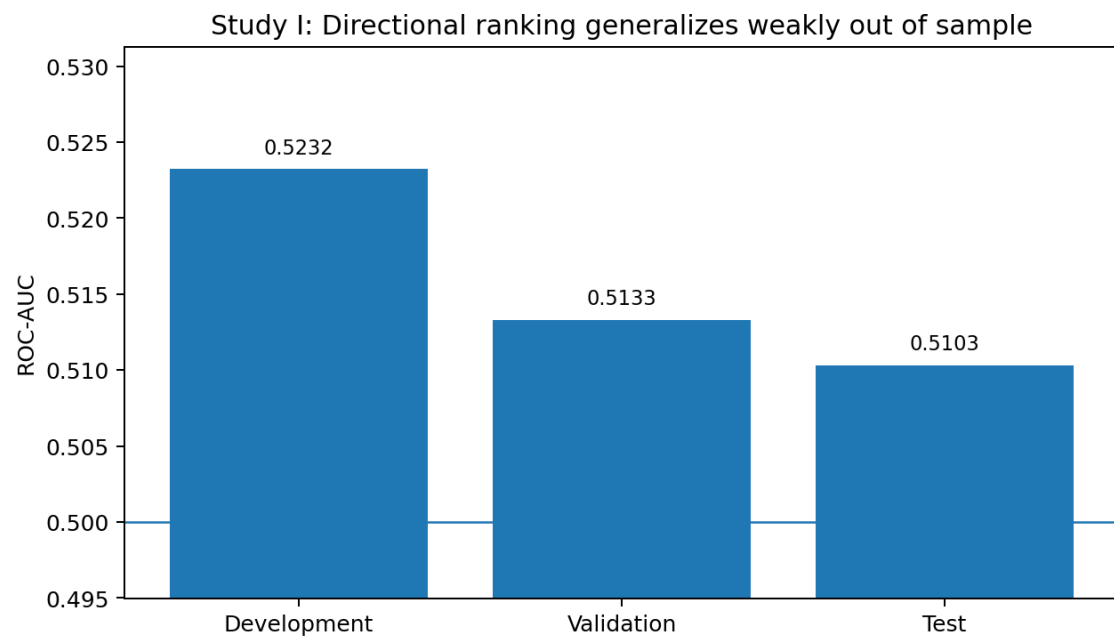


Figure 2. Directional ROC-AUC across development, validation and untouched test.

### 3. Study I-B - Economic validation

After statistical and predictive validation, the model specification was frozen and subjected to a separate economic protocol. November 2023 selected confidence thresholds; December independently confirmed or rejected each cost scenario. Only a December-passed scenario could open the later economic OOS.

#### Execution assumptions

| Component             | Frozen assumption               |
|-----------------------|---------------------------------|
| Decision              | Completed minute t features     |
| Entry                 | Reconstructed BBO at t+1        |
| Exit                  | Original t+10 forecast endpoint |
| Spread                | Crossed on both entry and exit  |
| Positions             | Non-overlapping                 |
| Extra round-trip cost | 0, 1, 2, 4, 6, 8, 10, 12 bps    |

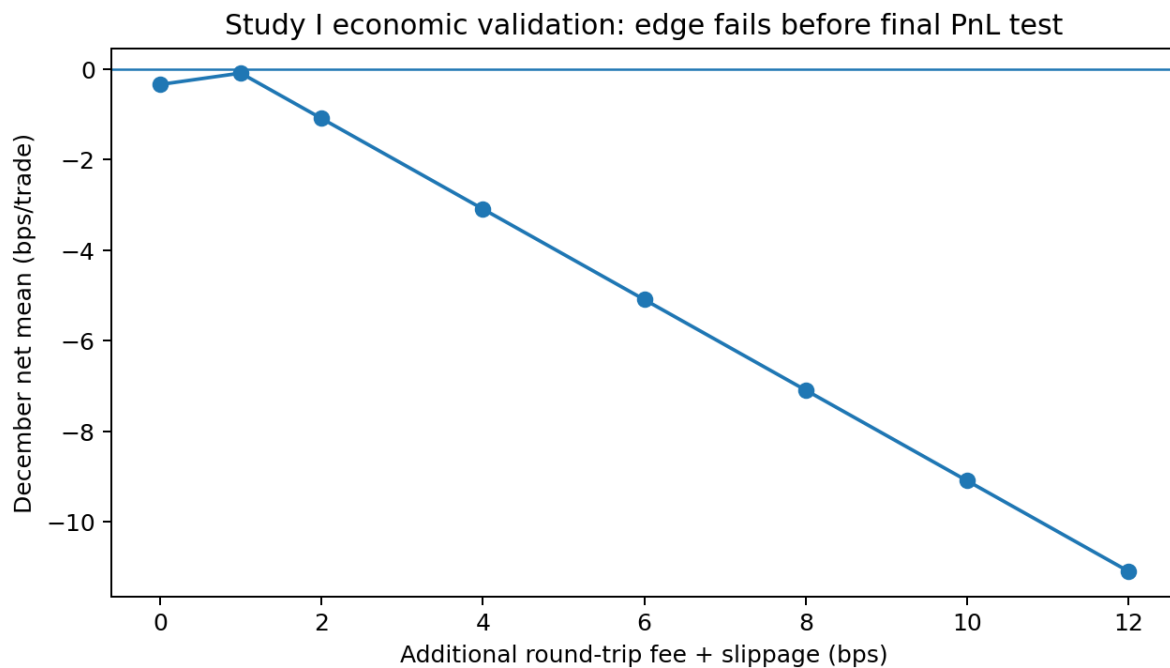


Figure 3. December net mean return after spread and additional cost.

Every scenario failed the December economic gate. At zero additional cost, the November-selected 50% coverage rule produced -0.3427 bps/trade in December. The 5% high-confidence threshold had positive gross December return (+0.9089 bps/trade), but its break-even additional round-trip cost was only 0.9089 bps; a 1-bp assumption already made mean net return negative.

**Decision: the Jan-Mar economic OOS was not opened. The correct conclusion is that the baseline contained statistical information but did not establish executable alpha under the conservative taker specification.**

## 4. Study II - Aggressive trade-flow extension

Instead of tuning the failed backtest, Study II introduced a genuinely new information source from Binance one-minute futures klines. For total volume  $V$  and taker-buy volume  $B$ , the one-minute trade-flow imbalance was defined as  $(2B - V) / V$ , with causal 5- and 10-minute volume-weighted aggregates.

The augmented logistic model improved ranking in both pre-OOS periods: +0.0042 AUC in November and +0.0060 in December. December's paired day-bootstrap Delta-AUC interval was entirely positive ([0.0005, 0.0116]).

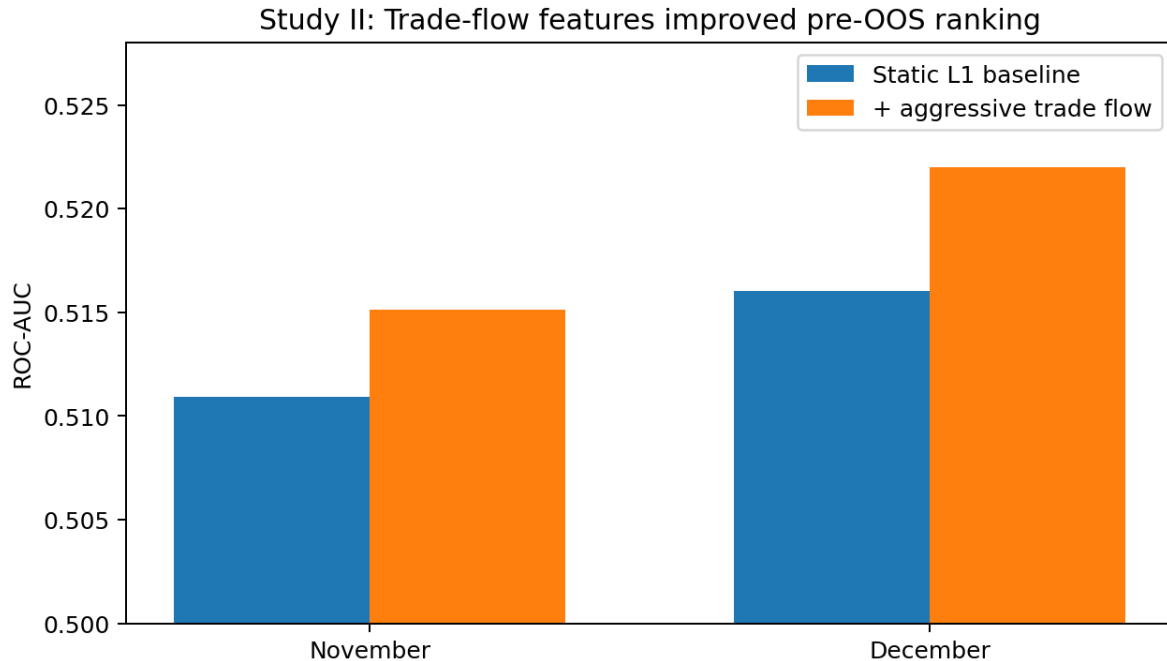


Figure 4. Static L1 baseline versus augmented trade-flow model.

However, the pre-registered continuation hypothesis was rejected. TFI coefficients were negative in every May-December 2023 month, and the largest augmented flow coefficients were negative. Aggressive buying pressure behaved as a short-horizon reversal signal rather than a continuation signal.

**Because the original sign gate required positive TFI behavior, the Study II gate correctly failed and Jan-Mar flow-enhanced OOS remained closed. The negative sign was preserved as a discovery to be tested on new data instead of being relabelled as success after the fact.**

## 5. Study III - Fresh 2025 reversal replication

Study III converted the unexpected negative TFI sign into a new hypothesis and moved to a genuinely newer calendar-year sample: 2025 Binance USD-M one-minute klines. The primary parameter-free score was -TFI\_5m, with -past-5-minute-return as a control for ordinary short-horizon mean reversion.

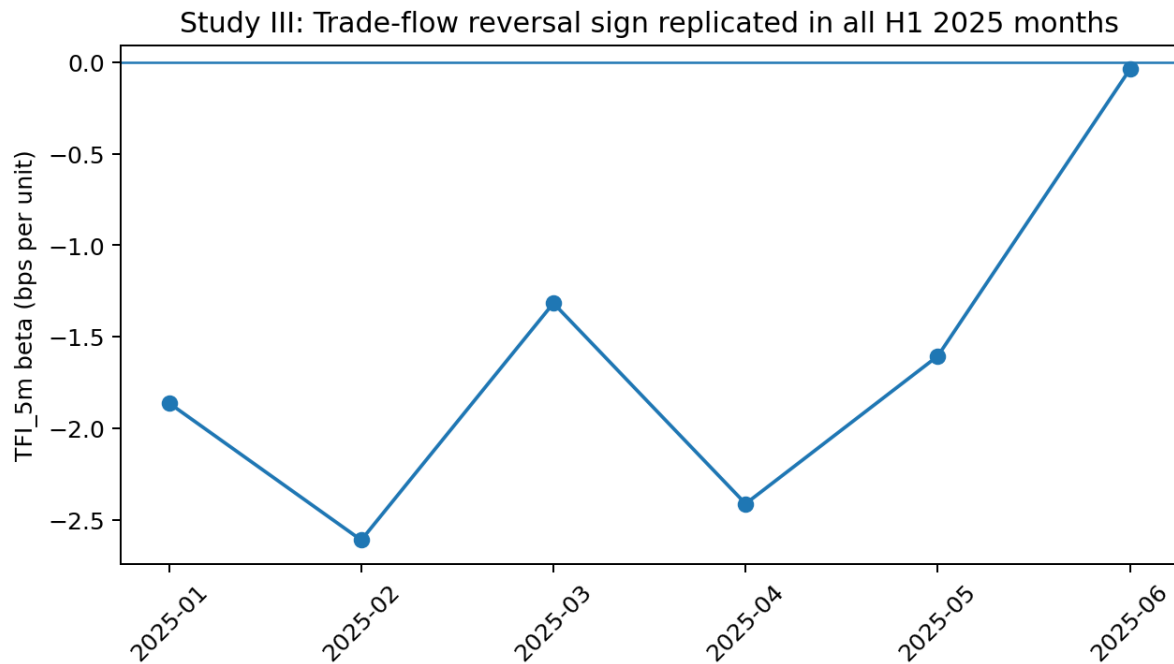


Figure 5. Monthly TFI\_5m coefficient, Jan-Jun 2025. All six months were negative.

| Period               | TFI beta         | HAC p   | -TFI AUC | -PastRet AUC |
|----------------------|------------------|---------|----------|--------------|
| Jan-Mar confirmation | -1.9028 bps/unit | 0.00003 | 0.5210   | 0.5176       |
| Apr-Jun replication  | -1.2810 bps/unit | 0.00014 | 0.5151   | 0.5142       |

The reversal sign replicated strongly. But the stronger incremental claim did not. After controlling for previous 5-minute price return, the Apr-Jun TFI coefficient remained negative but had HAC  $p=0.2244$ . Flow-reversal AUC exceeded simple price-reversal AUC by only  $+0.0008$ , below the frozen  $+0.001$  criterion, with a bootstrap interval spanning zero.

**Decision: the Study III gate failed 6/8 rather than being redefined after the result. Jul-Dec 2025 final OOS remained closed.**

## 6. Conclusions, limitations and portfolio relevance

### Empirical conclusions

| Claim                                             | Evidence                                | Status        |
|---------------------------------------------------|-----------------------------------------|---------------|
| Static book imbalance has information             | Stable positive beta; AUC > 0.5 OOS     | Supported     |
| Static-book model is executable alpha             | Independent economic gate               | Not supported |
| Aggressive TFI predicts continuation              | Pre-registered sign test                | Rejected      |
| Aggressive TFI predicts reversal                  | 2023 discovery + fresh 2025 replication | Supported     |
| TFI adds robust information beyond price reversal | Partial regression + Delta AUC gate     | Not supported |

### Limitations

The study covers one crypto instrument/venue and mostly one-minute aggregation. Queue position, true event-level OFI, funding, market impact, capacity, exchange outages and sub-minute latency are not fully modeled. Study III uses close-to-close returns because reliable later bookTicker coverage was unavailable. The economic backtest is a conservative research simulator, not a production execution engine.

### Why this is a quantitative-research portfolio project

The strongest portfolio signal is not a headline Sharpe ratio. It is the workflow: data integrity checks, exact-horizon target construction, robust inference, chronology, pre-specified gates, test-set restraint, execution-cost modelling, rejected hypotheses, and independent replication.

**Final takeaway: A small predictive edge can be statistically real, generalize weakly, and still be economically unusable. A surprising signal can replicate across years and still fail an incremental-information gate. Preserving those distinctions is the core research result.**

### References

Cont, R., Kukanov, A., & Stoikov, S. *The Price Impact of Order Book Events*.  
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